A MORPHISM CARRYING A RELATIVELY VERY AMPLE SHEAF IS SEPARATED

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Separatedness is local on the base. Schemes, Section \ref{schemes-section-separation-axioms}, cited at proof lines 9136-9138. Reduces the claim to the local situation.
- An immersion is separated, and the homogeneous spectrum of a graded ring is separated. Schemes, Lemma \ref{schemes-lemma-immersions-monomorphisms} and Constructions, Lemma \ref{constructions-lemma-proj-separated}, cited at proof lines 9139-9143. Give separatedness of the two maps into which the local morphism factors.

Source reference. Original lemma 01VQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2455–2455. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 01KH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2050–2050. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01L7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2076–2076. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01MC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2117–2117. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. Let $\mathcal{L}$ be an invertible sheaf on X. If $\mathcal{L}$ is relatively very ample on X/S then f is separated.*

Proof. Being separated is local on the base (see Schemes, Section 01KH). An immersion is separated (see Schemes, Lemma 01L7). Hence the lemma follows since locally X has an immersion into the homogeneous spectrum of a graded ring which is separated, see Constructions, Lemma 01MC. □